

Food for the fascia: Molecular and biochemical processes

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How to puzzle a fascia

Within the living myofascial system, some pieces are always degraded and some others are newly built all the time. Beside the inherent genetic and structural pattern, environmental conditions can have a huge influence. Factors such as food and nutrition, shear forces and mechanical overload have strong effects on quality and fine tuning of tissue puzzling.

The characteristics of the soft tissue are regulated according to the individual requirements. External stimuli help to create the structural details of the required synthesis with tailor-made adjustments. There are a number of further checkpoints, including the availability of all essential building materials, sufficient energy for the work of synthesis, turnover from restructuring and breakdown, as well as the activities of the stress axis with signal effects on post-translational epigenetic fine tuning.

The tissue's capacity for regulation, elasticity, stress and shear-force tolerance, deformation capacity, resilience, inflammatory status, homeostasis, dynamic reciprocity, capacity for recycling and new synthesis, are all critical parameters for human health. They are also major criteria for the strategy and success of each therapist, doctor, trainer, bodyworker and dancer working in this dynamic puzzle, and therefore concern areas of pain, the locomotor system, regeneration and the immune response.

The detailed biochemistry of our internal architecture offers far-reaching insight into a deeper understanding of the function, quality and dysregulation of fascia. This backdrop also points the way towards developing differentiated concepts for the rational and physiological nourishment of connective tissue structures.

The essential ingredients for high-performing fascia can be readily understood based on the principles of anatomy and biochemistry. Basically, any tissue in the body consists of two main components: cells and the extracellular matrix (ECM). While the cells are responsible for the synthesis of the ECM, the latter itself consists of fibers (providing tensile strength), water and ground substance. Primarily, ground substance is built up of water, galactose, mannose, glucosamine, omega-3 and amino acids.

There are several specific amino acids that are required for the synthesis of the large protein family of collagens. Additional requirements include vitamins B, C and D, plus minerals such as magnesium, calcium, manganese, zinc, selenium, copper, sulfur, silicon and phosphorus, as well as secondary plant substances.

The ground substance

The ground substance and its milieu are existential requirements for life with respect to nutrition, control, signaling activity, and the basic nature of cellular processes, both physiological

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and pathological. Thus, extracellular processes affect and control the cells themselves through contact with the cell surface, ligands and biochemical pacemaker processes. The intercellular substance is a critical element for understanding bacterial and viral infections, the pathways of carcinogenesis and metastasis, inflammatory processes, and neurodegenerative changes. Beginning from the earliest developmental stages of the central nervous system, structures of the extracellular matrix are essential for maturation, growth, migration and cell control, differentiation of neurons and glial cells, and organization of the connective tissues and fasciae.

In addition, the activity and complex regulatory functions of nerve cell growth factors, target-region dependent growth factors, and growth factors for collagen would not be possible without the fundamental functions of the substances located in the extracellular matrix.

Special features of the extracellular space

The ground substance is made up of a jelly-like substance that holds cells together and provides pathways for the diffusion and transport of nutrients, signal carriers, oxygen and messenger substances. It consists of an interwoven network of proteins and monosaccharides, which are called proteoglycans and glycosaminoglycans (Lindahl et al., 2017).

Looking at their architecture and micro-structure, these components are sponge-like, consisting of many branches, looking like fern or fir branches. They are imbued of water, hydrogen bonds and minerals. They provide resistance against different types of compression, shear forces, shock absorbance, elasticity, as well as tear strength. These qualities

are guaranteed by special networks, in which monosaccharides and proteins are connected and interwoven. Typical examples include proteins such as collagen, elastin, fibronectin, syndecans and the glycocalyx. The glycocalyx is the frosted-like surface of all cells.

All of the synthetic pathways for these sugar-amino-sugar molecules (glycosaminoglycans and proteoglycans) depend upon galactose (Lehninger et al., 2005). As a monosaccharide, galactose is the key substance, the building block, the dynamic adhesive and the key puzzler inside the surface antenna of all compounds within the ground substance, providing the signaling properties at the surface of all cells (see Figure 29.1). The surface of the cell can be seen as an interface and field of micro-antennas that are essential for all processes of cell to cell communication (Bibb and Nestler, 2006).

The process of how monosaccharides are combined to each other is akin to putting the pieces of a puzzle together, and is called glycosylation. The synthetic pathway also represents a critical biochemical genetic regulatory mechanism. This process involves large numbers of essential substrates, hormones, enzymes, messenger substances, receptors and life-sustaining components of the intra- and extra-cellular space. In conjunction with signal communication and cell recognition, monosaccharides like galactose play a vital role in cell adhesion, cell signal cascades, immune regulation, neuroregeneration and neurobiochemical processes.

Beside galactose, sulfate groups and sulfur-containing amino acids such as cysteine and methionine are essential key factors.

The three-dimensional (3D) network in soft tissue is designed to regulate metabolic acidosis and buffer pH levels.

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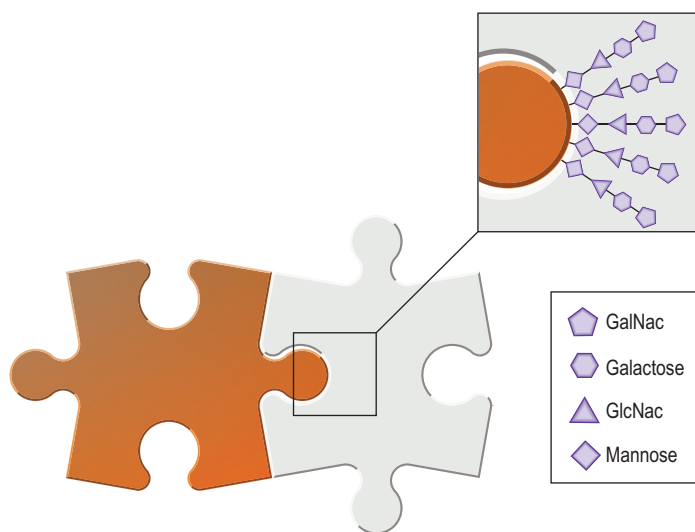


Figure 29.1

In the picture of the puzzle we talk about the arms and hands of the puzzle pieces. The biochemical metabolic pathways for galactose and mannose show how these terminal components of all heteroglycans or puzzle pieces represent the substrate for cellular biosynthesis. They act as metabolic and functional cell-to-cell anchors and communicators in the intercellular space of the ground substance.

Typical examples of glycosaminoglycans include hyaluronan, which serves as a lubricant in joint synovial fluid. In addition, their interactions with the extracellular matrix assure tensile strength and elasticity in cartilage, tendons and connective tissue.

Chondroitin sulfate is responsible for the tensile strength of cartilage, tendons and connective tissue. Dermatan sulfate assures the suppleness of skin, vessels and heart valves. Keratin sulfate is found in cartilage and bone as well as in the cornea and the nails. In the extracellular matrix, the network of proteoglycans and glycosaminoglycans mature to become a functional series of small molecular cytokines, growth factors and chemokines. They reside in the extracellular matrix and contribute to its spatial 3D structure while also serving as a standby reservoir in the proteoglycan receptors of syndecans. These components are highly dynamic, depend on the rule of offer and demand, and can be regulated by functional changes and requests. Therapists in the fields of osteopathy, yoga and myoreflex therapy, as well as teachers of Pilates,

Thai Chi and Qi Gong, claim to observe changes in these fields by activating processes of regulation within the ground substance.

These structures assure super-fast information processing along with fine regulation of the ground substance and endocrine control of cell response.

Glycosylation helps to determine almost all of the fine structures and functions of living tissues. Glycosylated proteins and lipids are essential structural components of cell membranes and help to shape cell interactions. Thus, glycosylation presents enormous biological storage capacity, greatly exceeding that of classical information carriers such as nucleic acids and proteins.

Non-disrupted biological glycosylation ensures physiological folding, functional protein conformation, seamless intracellular transport (protein targeting) and exocytosis, the stability of lipids and proteins, affinity for binding partners (such as the insulin receptor), and protects from proteolytic breakdown.

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“Together, the extracellular matrix and cells form a viscoelastic system, which maintains a self-stabilizing structure when impacted by external forces (‘tensegrity’). Therefore, the extracellular matrix represents an attractor for all external and internally acting forces similar to a coupled spring, whereby small causes can have very large effects.” (Heine, 2005)

Inside the puzzle of sugar-amino-sugar molecules building the ground substance, the family of fibroblast growth factors (FGF) are the clue in between the pieces of the puzzle (Landreth, 2006). To understand the activity of fibroblasts and to balance their physiological activity, nutritional factors and exercise are key factors.

The ground substance is also populated by cytokine families. “Sleeping” within this ground substance are myoblasts, fibroblasts, glial cells, chondroblasts, osteoblasts and mast cells, all of which are activated, transported and regulated according to need in building a fitting picture. There is a complex balance between situationally directed tissue restructuring, on one side by potent inflammatory substances, and on the other side by growth factors, including transforming growth factor beta (TGF- β) and nerve growth factor.

The large group of TGF- β growth factors regulates the synthesis of ground substance proteins, stimulating the breakdown of bone, connective tissue, muscle and the skin. They are essential for the stability of the bone matrix as well as for wound healing.

Immobilization and inflammation caused by junk food may result in impaired growth, bone growth disorders, chronic inflammation and infections, and can even cause the malignant transformation of cells.

In this setting, pro-inflammatory messenger substances such as prostaglandins, histamine, proteases and leukotrienes are key factors for

understanding a wide range of disease states. These processes can be balanced by training and exercise.

The ground substance and signal control

Integrins function as multidirectional nodal points and modify their docking behavior inside the soft tissue. Indeed, they look like the telescopes of a submarine. They communicate actively in all directions: similar to a mobile phone, they are able to receive, send and modulate messages, and can be online in the body network all the time.

The interaction of cells and their surface structures with components of the extracellular space has a profound impact on the development and behavior of individual cells.

Signals related to mechanical stress and associated burdens from the outside shape the living world of this ground substance. Depending on the nature of the signals, very different forms and types of connective tissue can be synthesized. Depending on use and activation, these signals will induce production of collagen and elastin or the synthesis of myofibroblasts. Even those structures that appear more solid and self-contained, such as bones and joints, are thus highly dependent upon function at the cellular level. And even in the context of severe developmental restrictions and dysplasia, they remain shapeable and modifiable well into advanced age. Many years of therapeutic experience and clinical progress show that myofascial back pain is often related to changes like edema, swollen and thickened areas, and inflammation in the thoracolumbar fascia. Different functional demands caused by active resistance stretch training (Mosetter and Mosetter, 2016; Mosetter 2017, 2018) induces different signals through transduction of a wide

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spectrum of glycoproteins. These are able to regulate the synthesis of the type of collagen. Inflammatory parameters decrease and elasticity is regained. In systemic sclerosis, scleroderma and lupus, stretching leads to anti-inflammatory activities, changing the composition of soft tissue (Berrueta et al., 2016; Xiong et al., 2017). In osteogenesis imperfecta, an active stretch with KiD and intense training on a vibrating platform (Galileo) are translated through glycoprotein signals in soft tissue and bone remodeling. Even the inner structure of the bones is changed in 3D architecture (Sa-Caputo et al., 2017). Density in corticalis is reduced to almost normal conditions and inner stability in inner architecture is switched on. Skin and soft tissue remodeling and recovery are also translated by these cross-talks.

The form, development, growth, polarization, movement and, at times, the nature of individual cells and their functions typically reflect corresponding shifts in the cellular environment.

Interactions and water binding between glycosaminoglycans and proteoglycans also generate fluctuations in charge and electrical and chemical gradients, which open information bridges and transport pathways. This mobility of the ground substance helps to maintain the proper organization of connective tissues, fascia, the cytoskeleton, and biochemical communication cascades.

The synthesis of connective tissue

Collagen is synthesized in a series of multiple intracellular and extracellular steps (Figure 29.2).

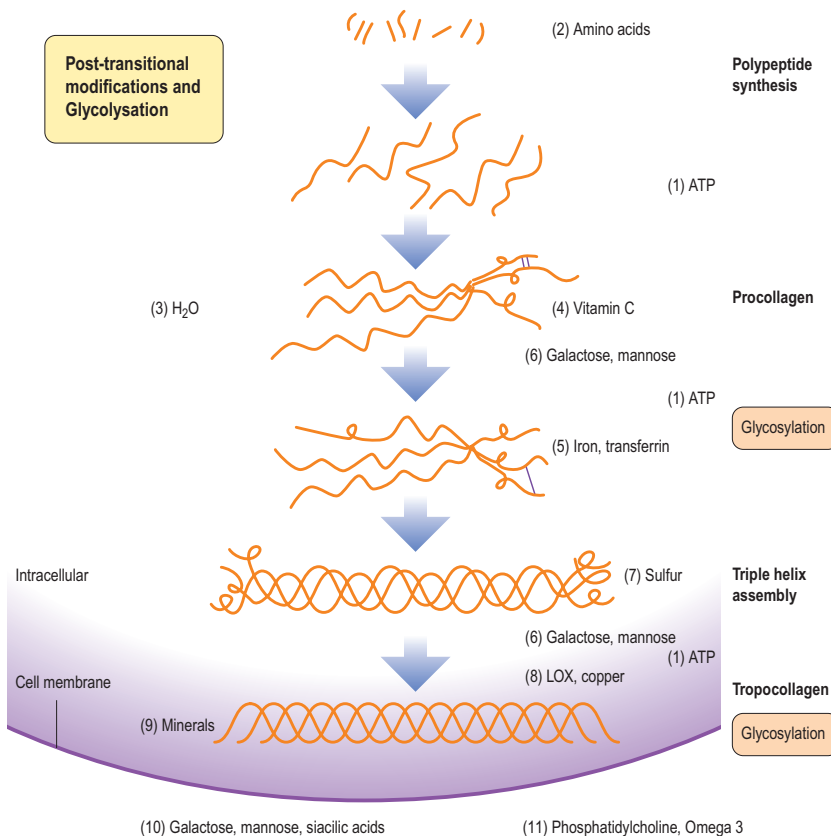


Figure 29.2

The steps of synthesis of connective tissue. Basic prerequisites, explanations and numbers in the text.

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Basic prerequisites

1: ATP

In each step of synthesis, turnover and repair of collagens are dependent on energy in the form of ATP, which is produced in the mitochondria. The more economical the function of mitochondria is, the more ATP is available, and then less metabolic acidosis, oxidative stress and inflammatory signals disturb the process of repair and regeneration.

2: Amino acids and proteins

Amino acids are essential block substances for all collagens and proteoglycans. In particular, lysine, proline, hydroxyproline, glycine, glutamine, GABA, β -leucine and arginine are very important. Deficiencies in the supply of amino acids impair physiological synthesis.

The individual amino acids are configured like pearls into a chain. Folding, which consumes ATP, results in the formation of 3D figures, spirals and custom-configured complex, helix-shaped structures. Through specific charging of the amino acid bonds and cross-linked attractive and repulsive forces, these pro-collagens are hardened, gaining tensile strength to become stabilized into stiff or elastic fibers.

3: Water and hydrogen bonds

Water molecules are an important component of this network system, with their specific lattice structures, their interconnections through hydrogen bonds, and their bridge formations with ions and minerals. The best nutritional source is still water (CO_2 in the water may disturb the processes because organic acids enforce metabolic acidosis).

4: Vitamin C

To create fibrillar collagen 1, which is responsible for the strength of bones, cartilage, tendons, ligaments, fascia, the dermis, sclerae, cornea and dentine, proline and lysine are hydroxylated with the assistance of vitamin C to hydroxyproline and hydroxylysine. Good sources are fruits low in glucose and fructose and high in fibers, such as berries, lemons, apricots, the old sort of apples, and papaya.

5: Iron and transferrin

Iron ensures the hydroxylation of proline, which is essential for stabilization of the tropocollagen spirals and triple helices. High concentrations of iron are found in organic, unprocessed red meat, fish, wholegrain products, cacao, nuts, almonds, sesame and poppy seeds.

6: Galactose and mannose

The hydroxylation of lysine is a precondition for the important steps that follow: through the adherence of monosaccharides, signal antennae are configured for task specificity, lifespan, contact behavior and 3D structure, etc. This critical process is known as glycosylation. Fine tuning takes place by means of building, dismantling and reconfiguring these small components by means of glycosyl transferases.

In the steps for fine adjustment of collagen to the milieu conditions of the microenvironment, the collage is tuned by the amino acids asparagine, serine, lysine and threonine with monosaccharides such as mannose and galactose.

Through the aggregation of several spirals, in the final intracellular step, complex triple-helix shapes are built with disulfide bridges.

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High amounts of galactose and mannose are found in lentils, blackberries and vegetables.

7: Sulfur-containing amino acids

For the next step in the synthesis, through the appendage of sulfur (sulfating), secretion into the extracellular space is initiated. After secretion, the propeptides are split off by small shears (peptidases). In this way, water-insoluble tropocollagens are created. The stringing together of different tropocollagen units ensures the formation of microfibrils, which ultimately are built up into fibrils by way of fibrillar collagen (Lindahl et al., 2017).

Eggs, fish, red organic meat, cashew, almonds, beans, chickpeas, lentils, pecorino and millet are all good sources.

8: Copper

Lysyl oxidase (LOX) is a copper-dependent enzyme that initiates wiring and crosslinking of elastin, fibronectin and collagen in the ground substance. LOX activity can be regulated by secondary plant substance, green tea, vitamin D, beta-carotene, vitamin B6 and *Calendula officinalis* (Kothapalli and Ramamurthi, 2009).

Good sources include artichokes, sprouts, cacao, cashew, lentils, beans and chickpeas.

9: Minerals

Minerals such as magnesium, manganese, copper, iron and zinc always act in their ionic-colloidal form within the gel-like network of charges; in fact, this is generally the only form in which they can be utilized by the body.

Calcium and magnesium are particularly important for bone formation, but phosphate,

arsenate, vanadate and silicate are also built into the corresponding collagen structures.

Salads, herbs, celery, vegetables, red and black rice, nuts, peas, lentils, oatmeal, sesame and wholegrains are all good sources.

10: Sulfur, galactose, mannose, charging the GAG

Extracellular proteins typically occur in glycosylated form. Glycans, typically hydrophilic and acidic from the negatively charged sialic acids, contribute substantially to the water solubility of proteins.

Eggs, fish and organic red meat contain sulfur. Still water as well as cranberry, goji berries and lentils support these processes.

11: Omega-3 fatty acids and phosphatidylcholine

Intracellular transport pathways across the dynamic border-layers of the cell membrane (via diffusion, active transport, vesicles, tubules and rafts) and normal contact behavior (adhesion, deformability) to the cytoskeleton require several additional substances, primarily omega-3 fatty acids, vitamin D and phosphatidylcholine/serine.

Good sources are wild deepwater fishes, seaweed, chia seeds, hemp oil, linseed oil and cod liver oil.

The enemies of ground substance

Mechanical overload and pain

Asymmetrical muscular-biomechanical leverage effects in fascia lead to corresponding chronic

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overloads. Along with the non-physiological forces, this causes metabolic, mechanical and molecular restructuring of the connective tissue with loss of function, resulting in pain (Eekhoff et al., 2018).

The ground substance of the fascia, with its glycosaminoglycans, is filled with free nerve endings, and has specific pain receptors for substance P and, in addition, is extensively innervated by branch connections from the sympathetic nervous system.

The ensuing problems include myofascial trigger points, hardened muscles and increased density of the fascial structures with a loss of elasticity. Fibroblasts, an especially agile cell type, respond to mechanical, molecular and metabolic stress with edema, synthesize smaller amounts of elastic collagen 1 and larger amounts of dysfunctional collagen 3, as well as a series of inflammatory mediators, including TGF- β 1, IL-1 β , IL-6 and TNF- α , thereby charting the course to fibrosis. Small blood vessels and nerve plexuses arise within the points of intersection in the fascia under shear stress. Vasoconstriction, ischemia, activation of pain messenger substances, and over-reactivity of the sympathetic system with neural pain sensitization, aggravate the crisis. Conditions of insulin resistance, metabolic acidosis, and low levels of coenzyme Q10, NADH, vitamin B3 and ATP, may all impair the rate of synthesis and the ultimate quality of collagen connective tissue and GAGs.

The flexibility from dynamic tensegrity, with its equilibrated tension-band system, becomes progressively impaired. Inflammation, excessive acidity, loss of water, slagging (reaction remains with deposit of waste, inflamed proteins which destroy the sponge of ECM), mineral and amino acid deficiencies, over time lead to coarse inflammatory thickening of the tissues, and fibrosis,

the deposition of advanced glycation end products (AGE) with pathological saccharification, oxidative stress and pain-generating excesses of ammonia, spread out through the connective tissue matrix.

The overconsumption of sugar

It is now generally agreed that the most destructive enemies of connective tissue are sugar, short-chain carbohydrates and trans fats (DiMauro and De Vito, 2006). Unphysiological excesses of sugar lead to sugar being burned with proteins in the process known as the Maillard reaction, and to increased production of AGE along with free radicals, which cancel out protein synthesis. The levels of glycine, proline, lysine, glutamine, GABA and acetylcholine fall.

As the causative agent for bad “aging” and diseases of civilization, one significant mechanism is a biochemical reaction which was already understood in 1912. When sugar reacts with proteins, so-called “glycation” products are produced. The biochemist Louis Maillard described the precise process by which sugar and milk proteins caramelize and burn in the cooking pot. Too much sugar in the blood sticks together and caramelizes not only hemoglobin in the red blood cells, but also nerves, blood vessels, essential fats, messenger substances, signal proteins, and hormones as well as their docking sites, nucleic acids, the extracellular matrix, and the connective tissues.

The larger the quantity of caramelized glycation products that are produced, the more widespread are advanced, complex burnt AGE. This preventable escalating metabolic process represents a catastrophe for the entire body and all of its cells (Ott et al., 2014).

The damage runs even deeper. The increased AGE lead to a rise in the corresponding binding

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sites and receptors for AGE, known as RAGE. Through these receptors, sabotage signal cascades are initiated in the interior of cells, including abnormal cell nuclear activation. Oxidative stress, mitochondrial dysfunction, disorders of the autophagosomes and degenerative processes are the biological consequences (Chen et al., 2017; Peng et al., 2018).

AGE significantly induce macrophages to express IL-6 and TNF- α . M1 macrophage markers such as iNOS and surface markers are significantly upregulated (while M2 macrophage markers such as Arg1 and CD206 remain unchanged after AGE stimulation.) AGE significantly increase RAGE expression in macrophages and activate NF- κ B pathway. Together with m-Tor signaling, the cytokines TGF-1 β , IL-1 β , IL-6 and TNF- α are all chronically upregulated (Derk et al., 2018). Post-translational modification and glycosylation are severely undermined by AGE!

The overconsumption of short-chain carbohydrates and sugar, with associated hyperglycemic states with AGE, insulin resistance conditions with oxidative stress, ATP deficiency, shortage of energy, metabolic acidosis, toxic breakdown products, excessive ammonia, chronic inflammatory changes, as well as crisis-dependent insulin and AGE/RAGE signal cascades, all lead to degeneration of connective tissue along with the related cell groups (Bierhaus, 2004; Chang et al., 2014; Serban et al., 2015)

At the same time, disruptions of energy metabolism, stress metabolism and states of emergency lead to the interruption of all regenerative metabolic pathways. The pathways for repair and synthesis of fascia, connective tissue, skin and neuronal repair are all inhibited. The consequences are injuries to the fascia, ligaments, muscle and joints. Painful muscles, fascia, and joints, inflammatory swelling of the connective

tissue, fibrotic degeneration, striae, stress fractures, rheumatic conditions, osteoporosis, and periosteal pain may all be sequelae of malnutrition and a deficient diet. Chronic activation of the sympathetic nervous system in the terminal portion of the extracellular matrix and the myofascial connective tissues intensify the spiral of pain and inflammation. Increasing levels of TNF- α , TGF-1 β , IL-1 β and IL-6 all culminate in degenerative changes.

Insulin resistance and hyperglycemia also inhibit physiological glycosylation, and, additionally, expose the soft tissues to stress-associated modifications (O-glycosidic acetylation, O-GlcNac). These processes set the stage for degeneration and block natural autophagosomes (Zhu et al., 2018).

What to do? An overview

The elimination of toxic overloads and nutritional medical care by means of natural eating, in particular, a glycoplan (Mosetter, 2018) and ketogenic diet (Wilson et al., 2017), can set in motion turning points for regeneration. Enhancing energy metabolism, especially in those hubs of ATP metabolism, the mitochondria, boosts the processes of collagen synthesis, rapidly synchronized protein folding, functional diversity, and—of critical importance in this context—their regular breakdown, economical digestion and tailor-made recycling.

Helpful essential replacement therapy with NADH, coenzyme Q10, ribose and creatine prepare the organism for a new start. They promote elimination and digestion of waste, while at the same time they can also facilitate the start of reparative processes and new synthesis. Glucosamine and galactose are first aid instruments in the repair toolkit and provide critical assistance.

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Endogenous detoxification

Irreversible ammonia production that occurs in stress metabolism, and causes damage to ion channels, receptors and nerve cells, can be redirected along an alternative metabolic pathway (Butterworth, 2006).

Amino acids are formed with galactose from toxic metabolites and ammonia as well as ammonia equivalents. In this process, detoxification takes place in the form of endogenous detoxification with a recycling feature. Moreover, mediators that are required for normal brain function can be synthesized from the amino acids (Mosetter, 2018). The body's endogenous detoxification steps can benefit from the broad activities of galactose through so-called pump glycoproteins with respect to structure, recognition and ATP-dependent performance.

Anabolic metabolism

The situation of catabolic degradative metabolism of protein with amino acid loss can be categorically diverted into constructive anabolic pathways.

In the metabolism of ammonia and ammonia equivalents, amino acids are produced from a portion of the galactose. This process takes up toxic ammonia produced by the stressed organism (in stress metabolism). Thus, detoxification is achieved in this simple protective pathway. The amino acids formed in this way can serve in the formation of valuable proteins. By metabolizing ammonia, galactose ensures the synthesis of alpha-ketoglutarate, glutamic acids, glutamate and aspartate (Yudkoff, 2006). In situations of insulin resistance, galactose can induce the activity of osteoblasts.

Supplementing the anti-oxidative protection program with zinc, selenium, magnesium, alpha

lipoic acid, ribose and glutathione further aids these measures.

Phosphatidylcholine and omega-3 fatty acids are utilized in the regenerative processes in the cell membrane.

The individualized rebalancing of deficient amino acid levels using a broad spectrum of essential amino acids (glycine, proline, leucine, lysine, glutamine, GABA) provides the raw materials for potential new synthesis.

The sulfur-containing amino acids methionine and cysteine have anti-oxidative properties while the sulfate groups are also co-factors in collagen synthesis.

Vitamin C, vitamin D, basic minerals, and metals, particularly iron, copper, zinc, magnesium and selenium, manganese, calcium, silicon and phosphorus, ensure good nutrition and energy supply of the fascia and also ensure that they do not become matted.

Provision of the monosaccharides, galactose, mannose and sialic acid, along with glucosamine, facilitates essential metabolic pathways of glycosylation (see above).

Omega-3 fatty acids and the spectrum of omega-3 fatty acids also bring into play so-called "resolving factors" (Zhang et al., 2017).

The 2016 Nobel Prize in Medicine has underscored the essential role of the mitochondria for successful autophagy, and proves that the activation of mitochondria, intermittent fasting, ketogenic diet, natural eating with the glycoplan and moderate training can fundamentally promote these processes.

The large family of myokines is capable of regulating fibroblasts as well as the structure of connective tissue. Stroma, cartilage and bone metabolism may thereby be reset towards a

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reparative function. Fibroblast growth factor-21 (FGF-21), BDNF, IL-15, IL-6 and IL-1 β are associated with increased soft tissue repair (Kaji, 2018; Guo et al., 2017).

Important actuators of successful repair are activated through the bowel and the microbiome.

Using prebiotics and probiotics, through the successful metabolism of dietary fiber and superfoods, well-synchronized gut bacteria can synthesize short chain fatty acids, which are key substances for intestinal health, the immune system, and also for the metabolism of the fascia. In particular, the healthy bacteria *Faecalibacterium prausnitzii* and *Akkermansia muciniphila* support all manner of tissue repair, especially in the ECM. These factors include butyrate, folic acid, biotin, vitamin B3, propionate and acetate.

The 2016, 2017 and 2018 Nobel Prizes for Medicine highlight the potential for self-healing powers through intrinsic bodily repair mechanisms, that is, the essential role of the

mitochondria for successful autophagy, the activation of internal rhythmicity (mindfulness), as well as the reparative capacities of immune and regeneration metabolism. Intermittent fasting, a ketogenic diet, natural eating with the glycoplan, supplementation, KiD (strength through stretch training) and moderate training can fundamentally promote all reparative processes.

Major protective anti-inflammatory factors can also be activated through resistance and muscle-fascia length training (Bouffard et al., 2008; Berrueta et al., 2018; Guzzoni et al., 2018) and active resistance stretch training (Mosetter and Mosetter, 2016; Mosetter 2017, 2018).

The most recent insights of pioneers in the field of fascia research help to confirm experience, concepts and hypotheses regarding Yoga, Tai Chi, Qi Gong, Pilates and active resistance stretch training. Form follows function. The better you move, the less you will suffer from inflammation.

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